

# FAZIL FIROSE IBRAHIM

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## SUMMARY

Computer Science (Data Science) graduate with hands-on experience building AI-powered applications using Python, LLM APIs (Gemini, Groq), and full-stack frameworks including Flutter, Flask, and Node.js. Independently designed, built, and deployed two AI products — a document-grounded exam-prep assistant and a voice-enabled math solver — alongside academic projects spanning machine learning, mobile, and web development. Seeking entry-level Software Engineer, Python Developer, Data Analyst, or AI Engineer roles while actively expanding expertise in Generative AI, RAG pipelines, and AI agents.

## EDUCATION

### B.Tech in Computer Science (Data Science)

Oct 2022 – Jun 2026

SCMS School of Engineering and Technology, Angamaly | CGPA: 8.00 / 10

## EXPERIENCE

### Machine Learning Intern | ICT Academy

Jun 2024

- Completed a 1-week Machine Learning internship at ICT Academy, focused on applied ML fundamentals.
- Performed data preprocessing and cleaning on real-world datasets using Pandas.
- Built and evaluated supervised learning models using Scikit-learn.
- Applied classification and regression techniques to sample datasets under mentor guidance.

## PROJECTS

### AceKTU — AI-Powered Exam Prep Assistant (Python, Streamlit, Gemini API)

2026

- Built and deployed a study-planning web app for engineering students that generates day-by-day exam schedules from uploaded syllabi (PDF/PPTX).
- Integrated the Google Gemini API with search grounding and JSON-mode output for reliable, real-time syllabus analysis.
- Implemented live progress tracking with interactive Plotly donut charts and a persistent session-state architecture to resolve Streamlit widget-state bugs.
- Added quick-revision PDF export and invalid-subject-code detection with automatic retry logic; deployed on Streamlit Community Cloud.

### MathPilot AI — Voice-Enabled Math Solver (Python, Groq API, SymPy, CustomTkinter)

2026

- Developed a desktop math assistant combining natural-language understanding (Groq/Llama) with symbolic computation (SymPy) for equations, calculus, and simplification.
- Built a full voice interaction pipeline: speech recognition, a custom "Hey Math" wake-word, and text-to-speech via Edge-TTS.
- Designed a modular architecture with separate AI, solver, voice, and GUI layers, using a dark-themed CustomTkinter interface.

### AI-Based Customer Churn Prediction using CNN (Python, TensorFlow/Keras, Flask, Node.js)

Jan 2026

- Developed a Convolutional Neural Network to predict customer churn on a dataset of 10,000+ subscription customers.
- Performed data preprocessing, feature engineering, model training, and evaluation to improve prediction accuracy.
- Built a Flask + Node.js + Flutter web application to visualize predictions through an interactive interface.

### Bus Pass Management System (Flutter, Node.js, Firebase)

Dec 2024 – Apr 2025

- Developed a QR-code-based bus pass platform supporting three user roles: Admin, Student, and Conductor.
- Integrated the Razorpay payment gateway for secure online transactions.
- Implemented real-time QR validation and pass-status management.

### Grocery Delivery App (Flutter)

Oct 2024

- Built a grocery delivery application with user authentication, product browsing, cart management, and order placement.
- Implemented real-time order tracking and a secure payment workflow with a responsive mobile UI.

## SKILLS

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**Programming Languages:** Python, SQL, Java, C++, C, R

**AI / ML & Data Science:** Machine Learning, Generative AI, LLM APIs (Gemini, Groq), Pandas, NumPy, Scikit-learn, TensorFlow/Keras, SymPy, Data Analysis, Data Cleaning, Data Visualization, EDA

**Frameworks & Tools:** Flutter, Flask, Firebase, Node.js, Streamlit, Git, GitHub

**Currently Learning:** FastAPI, RAG Pipelines, AI Agents, PostgreSQL

## CERTIFICATIONS

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**GenAI Powered Data Analytics Job Simulation** — Forage (Tata) *(Jul 2026)*

- Completed practical tasks in exploratory data analysis, AI-driven delinquency prediction, and data storytelling for a collections strategy.

**Advanced SQL** — Kaggle *(Jul 2026)*

**Python** — Kaggle *(Jul 2026)*

**Python for Data Science** — Infosys Springboard *(2025)*

- Covered Python programming, NumPy, Pandas, data preprocessing, and basic visualization techniques.

**Workshop on Full Stack Development (Angular)** — ICT Academy of Kerala *(Oct 2023)*

- 3-day hands-on workshop on building modern web applications using Angular.

**Workshop in Metaverse** — GEC Thrissur *(2023)*